TA Session

Topics: Sample Mean • Point Estimation • Sample Variance (Known and Unknown Population Mean)

Formula Sheet

1. Sample mean

$$\bar{x} = (\Sigma x_i) / n$$

For frequency data:

$$\bar{x} = (\Sigma f_i x_i) / (\Sigma f_i)$$

2. Point estimation

Population mean estimate: $\hat{\mu} = \bar{x}$

Population proportion estimate: $\hat{p} = x / n$

3. Population mean known — average squared deviation from the known mean:

$$s^2\mu = [\Sigma(x_i - \mu)^2] / n$$

4. Population mean unknown — unbiased sample variance:

$$s^2 = [\Sigma(x_i - \bar{x})^2] / (n - 1)$$

Computational formula:

$$s^2 = [\Sigma x_i^2 - n(\bar{x})^2] / (n - 1)$$

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SAMPLE MEAN

Question 1. Daily electricity consumption

A school administrator wants to estimate the average daily electricity consumption of the classrooms in a school. Instead of recording the consumption of every classroom, she selects six classrooms at random. Their daily electricity consumption, in kWh, is recorded as follows:

Classroom	Daily consumption (kWh)
A	18
B	22
C	16
D	25
E	21
F	14

Calculate the sample mean and explain what it tells the administrator about the school's classrooms.

Detailed solution

Step 1: Number of sampled classrooms n = 6; observations are 18, 22, 16, 25, 21 and 14.

Step 2: Apply the sample mean formula: $\bar{x} = \Sigma x_i / n$.

Step 3: $\Sigma x_i = 18 + 22 + 16 + 25 + 21 + 14 = 116$.

$\bar{x} = 116 / 6 = 19.33$

Final answer: The sample mean is approximately 19.33 kWh per classroom per day. It estimates the average daily electricity consumption of all classrooms in the school.

Question 2. Average marks with a missing observation

A teacher records the marks obtained by seven students in a mathematics test. The average marks of the seven students are calculated to be 72. However, one student's marks were accidentally omitted from the record. The six available marks are 65, 78, 71, 69, 80 and 74. Determine the missing student's marks and verify that the sample mean is 72.

Detailed solution

Step 1: Total marks of all seven students = $n \bar{x} = 7 \times 72 = 504$.

Step 2: Sum of the six known marks = 65 + 78 + 71 + 69 + 80 + 74 = 437.

Step 3: Let the missing mark be x. Then 437 + x = 504, so x = 67.

Step 4: Verify: $\bar{x} = (437 + 67) / 7 = 504 / 7 = 72$.

Final answer: The missing student's marks are 67.

Question 3. Sample mean from grouped frequency data

A factory manager studies the number of defective items found in a sample of production batches. The following table records the number of defective items per batch and the number of batches in each category. Calculate the mean number of defective items per batch in the sample. Based on this result, estimate the average number of defective items in a batch produced by the factory.

Defective items x_i	Number of batches f_i	$f_i x_i$
0	4	0
1	6	6

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Defective items x_i	Number of batches f_i	$f_i x_i$
2	8	16
3	5	15
4	2	8
Total	25	45

Detailed solution

For frequency data, $\bar{x} = \Sigma f_i x_i / \Sigma f_i$.

From the table, $\Sigma f_i = 25$ and $\Sigma f_i x_i = 45$.

$$\bar{x} = 45 / 25 = 1.8$$

Final answer: The sample mean is 1.8 defective items per batch. It is a point estimate of the population's average number of defective items per batch.

Question 4. Comparing the means of two samples

Two laboratories measure the time, in minutes, required to complete the same chemical analysis. Laboratory A records the following times for five randomly selected tests: 12, 15, 11, 14 and 18 minutes. Laboratory B records the times for six tests as 13, 12, 16, 15, 14 and 14 minutes. Calculate the sample mean for each laboratory and determine which laboratory has the lower average completion time in the samples.

Detailed solution

Laboratory A: $\bar{x}A = (12 + 15 + 11 + 14 + 18) / 5 = 70 / 5 = 14$ minutes.

Laboratory B: $\bar{x}B = (13 + 12 + 16 + 15 + 14 + 14) / 6 = 84 / 6 = 14$ minutes.

Final answer: Both laboratories have a sample mean completion time of 14 minutes. The samples do not show a difference in average completion time, although individual test times differ.

Question 5. Mean from a partially described sample

A researcher records the daily water consumption of eight households in a residential area. The average consumption is found to be 420 litres per household per day. Seven of the recorded values are 380, 450, 400, 500, 390, 430 and 410 litres. Find the missing household's consumption. If the researcher uses the sample mean to estimate the average consumption of all households in the area, what estimate will be reported?

Detailed solution

Step 1: Total consumption of eight households = $n \bar{x} = 8 \times 420 = 3360$ litres.

Step 2: Sum of the seven known observations = 380 + 450 + 400 + 500 + 390 + 430 + 410 = 2960 litres.

Step 3: Let the missing value be x. Then 2960 + x = 3360, so x = 400 litres.

Step 4: Verify: $\bar{x} = 3360 / 8 = 420$ litres.

Final answer: The missing consumption is 400 litres per day, and the estimated average for all households is 420 litres per household per day.

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POINT ESTIMATION

Question 6. Estimating the average lifetime of batteries

A manufacturer claims that its rechargeable batteries have a long operating life but wants an independent estimate of their average lifetime. A quality-control officer randomly selects five batteries and tests them until they stop functioning. Their lifetimes are 410, 435, 390, 450 and 415 charge cycles. Use the sample to obtain a point estimate of the population mean battery lifetime. Explain why the estimate is called a point estimate.

Detailed solution

Step 1: The population mean lifetime μ is unknown.

Step 2: Use the point estimator $\hat{\mu} = \bar{x} = \Sigma x_i / n$.

Step 3: $\Sigma x_i = 410 + 435 + 390 + 450 + 415 = 2100$.

$\hat{\mu} = \bar{x} = 2100 / 5 = 420$

Final answer: The point estimate of the population mean battery lifetime is 420 charge cycles. It is a point estimate because it gives one numerical value, rather than an interval, for the unknown population mean.

Question 7. Estimating the proportion of defective products

A packaging company wants to know whether its production line is meeting its quality standards. During a routine inspection, an inspector randomly selects 160 packaged units from a large day's production. On examination, 12 units are found to have damaged seals. Estimate the proportion of all units produced that have damaged seals. Also express the estimate as a percentage and state the estimated number of damaged units in a production run of 2,000 units.

Detailed solution

Step 1: Sample size $n = 160$ and number of defective units $x = 12$.

Step 2: Sample proportion $\hat{p} = x / n = 12 / 160 = 0.075$.

Step 3: Percentage estimate $= 0.075 \times 100 = 7.5\%$.

Step 4: Estimated damaged units in 2,000 units $= 2000 \times 0.075 = 150$.

Final answer: Point estimate = 0.075; defective percentage = 7.5%; estimated damaged units out of 2,000 = 150. This assumes the sample represents the production run reasonably well.

Question 8. Estimating population variance from a sample

A school wants to estimate the variability in the heights of its Class 12 students. A random sample of five students has heights of 158, 162, 160, 165 and 155 cm. The population mean height of all students is not known. Calculate the sample mean and use it to obtain the unbiased point estimate of the population variance.

Detailed solution

Step 1: $\bar{x} = (158 + 162 + 160 + 165 + 155) / 5 = 800 / 5 = 160$ cm.

Height x_i (cm)	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
158	−2	4
162	2	4
160	0	0
165	5	25
155	−5	25

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Height x_i (cm)	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
Total	0	58

Step 2: Use the unbiased sample variance formula $s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$.

$s^2 = 58 / (5 - 1) = 58 / 4 = 14.5$

Final answer: Sample mean = 160 cm; unbiased point estimate of population variance = 14.5 cm².

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SAMPLE VARIANCE WHEN THE POPULATION MEAN IS KNOWN

For these questions, the population mean is explicitly given. Calculate squared deviations from that known mean, not from the sample mean.

Question 9. Quality control in a filling plant

A beverage company has calibrated a machine so that the true long-run mean filling volume is known to be 500 ml. A technician selects five bottles from the production line and measures their contents: 498, 502, 497, 501 and 502 ml. Calculate the average squared deviation of the sampled volumes from the known population mean. Explain what the result measures.

Detailed solution

Step 1: Known population mean $\mu = 500$ ml; sample size $n = 5$.

Step 2: Use $s^2_\mu = \Sigma(x_i - \mu)^2 / n$.

x_i (ml)	$x_i - \mu$	$(x_i - \mu)^2$
498	−2	4
502	2	4
497	−3	9
501	1	1
502	2	4
Total	0	22

Step 3: Average squared deviation = 22 / 5 = 4.4 ml².

Final answer: The average squared deviation from the known population mean is 4.4 ml². It measures the squared spread of sampled filling volumes around 500 ml.

Question 10. Variability in examination marks

A large examination board has established that the population mean score for a particular standardized test is 70 marks. To examine the variation in a small group, an examiner selects six students whose scores are 65, 72, 68, 75, 70 and 66. Calculate the average squared deviation from the known population mean. Do not calculate the variance using the sample mean.

Detailed solution

Step 1: Known mean $\mu = 70$ and sample size $n = 6$.

Step 2: Use $s^2_\mu = \Sigma(x_i - \mu)^2 / n$.

Score x_i	$x_i - 70$	$(x_i - 70)^2$
65	−5	25
72	2	4
68	−2	4
75	5	25
70	0	0
66	−4	16
Total	−4	74

Step 3: Average squared deviation = 74 / 6 = 12.33 marks² (approximately).

Final answer: The average squared deviation from the known population mean is approximately 12.33 marks². The deviations do not sum to zero because the sample

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mean need not equal the known population mean.

Question 11. Checking the consistency of a weighing instrument

A laboratory uses a weighing instrument whose calibrated population mean error is known to be zero grams. During a test, six measurement errors are recorded as +1, −2, +3, 0, −1 and +2 grams. Calculate the average squared deviation of the errors from the known population mean. Based on this calculation, state the estimated root-mean-square deviation from the known mean.

Detailed solution

Step 1: Known mean $\mu = 0$ and $n = 6$.

Step 2: Use $s^2\mu = \Sigma(x_i - \mu)^2 / n$.

Error x_i (g)	$x_i - \mu$	$(x_i - \mu)^2$
+1	1	1
−2	−2	4
+3	3	9
0	0	0
−1	−1	1
+2	2	4
Total	3	19

Step 3: Average squared deviation = 19 / 6 = 3.167 g² (approximately).

Step 4: RMS deviation = $\sqrt{(19 / 6)} \approx 1.78$ g.

Final answer: Average squared deviation = 3.167 g²; root-mean-square deviation = 1.78 g.

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SAMPLE VARIANCE WHEN THE POPULATION MEAN IS UNKNOWN

Question 12. Variation in the time taken to solve a problem

A mathematics teacher wishes to study how consistently students solve a particular problem. The teacher randomly selects five students and records the time taken by each student: 8, 12, 10, 15 and 5 minutes. The average time for the entire class has not been determined. Calculate the sample mean and the unbiased estimate of the population variance of the solving times.

Detailed solution

Step 1: $\bar{x} = (8 + 12 + 10 + 15 + 5) / 5 = 50 / 5 = 10$ minutes.

Time x_i (min)	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
8	−2	4
12	2	4
10	0	0
15	5	25
5	−5	25
Total	0	58

Step 2: $s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1) = 58 / 4 = 14.5$.

Final answer: Sample mean = 10 minutes; unbiased estimate of population variance = 14.5 minutes².

Question 13. Calculating variance using a shortcut formula

A delivery company wants to understand the variation in delivery times for a particular route. A random sample of six deliveries has completion times of 20, 25, 30, 22, 28 and 25 minutes. The population mean delivery time is unknown. Find the sample mean and the unbiased sample variance. Use the computational formula for variance rather than calculating every deviation from the mean.

Detailed solution

Step 1: $\Sigma x_i = 20 + 25 + 30 + 22 + 28 + 25 = 150$; therefore $\bar{x} = 150 / 6 = 25$ minutes.

Step 2: $\Sigma x_i^2 = 20^2 + 25^2 + 30^2 + 22^2 + 28^2 + 25^2 = 400 + 625 + 900 + 484 + 784 + 625 = 3818$.

Step 3: Use $s^2 = [\Sigma x_i^2 - n(\bar{x})^2] / (n - 1)$.

$s^2 = [3818 - 6(25)^2] / 5 = (3818 - 3750) / 5 = 68 / 5 = 13.6$

Final answer: Sample mean = 25 minutes; unbiased sample variance = 13.6 minutes².

Question 14. Sample variance from grouped frequency data

A researcher studies the number of books read by students during a month. A random sample of students gives the following frequency distribution. The population mean number of books read is unknown. Calculate the sample mean and the unbiased estimate of the population variance.

Books x_i	Frequency f_i	$f_i x_i$	$f_i x_i^2$
1	2	2	2
2	4	8	16
3	3	9	27
4	1	4	16

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Books x_i	Frequency f_i	$f_i x_i$	$f_i x_i^2$
Total	10	23	61

Detailed solution

For frequency data, $n = \Sigma f_i = 10$; $\Sigma f_i x_i = 23$; $\Sigma f_i x_i^2 = 61$.

Step 1: Sample mean $\bar{x} = 23 / 10 = 2.3$ books.

Step 2: Use $s^2 = [\Sigma f_i x_i^2 - n(\bar{x})^2] / (n - 1)$.

$s^2 = [61 - 10(2.3)^2] / 9 = (61 - 52.9) / 9 = 8.1 / 9 = 0.9$

Final answer: Sample mean = 2.3 books; unbiased sample variance = 0.9 books².

Question 15. Difference between known and unknown means

A laboratory is investigating the variability of a chemical measurement. Six measurements, in arbitrary units, are obtained: 9, 11, 10, 12, 8 and 10. In an earlier calibration study, the laboratory established that the population mean for this measurement is 10 units. A student calculates the variance in two ways: (A) use the known population mean of 10 and divide the sum of squared deviations by n; (B) treat the population mean as unknown, estimate it from the sample, and use n − 1 in the denominator. Calculate the variance obtained by each method and explain why the answers are different.

Detailed solution

Step 1: Sample mean $\bar{x} = (9 + 11 + 10 + 12 + 8 + 10) / 6 = 60 / 6 = 10$. In this sample, the sample mean equals the known population mean.

x_i	$x_i - 10$	$(x_i - 10)^2$
9	−1	1
11	1	1
10	0	0
12	2	4
8	−2	4
10	0	0
Total	0	10

Step 2 — Method A, mean known: $s^2\mu = \Sigma(x_i - \mu)^2 / n = 10 / 6 = 1.667$ units².

Step 3 — Method B, mean unknown: $s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1) = 10 / 5 = 2$ units².

Final answer: Method A gives 1.667 units²; Method B gives 2 units². They differ because Method A averages squared deviations about a genuinely known mean, whereas Method B estimates the mean from the sample and uses n − 1 to obtain an unbiased estimator of population variance.

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QUICK REVISION — WHICH FORMULA SHOULD YOU USE?

Situation	Formula
Average of observations	$\bar{x} = \Sigma x_i / n$
Average from frequency table	$\bar{x} = \Sigma f_i x_i / \Sigma f_i$
Estimate unknown population mean	$\hat{\mu} = \bar{x}$
Estimate population proportion	$\hat{p} = x / n$
Known population mean; average squared deviation	$\Sigma(x_i - \mu)^2 / n$
Unknown population mean; unbiased variance estimate	$\Sigma(x_i - \bar{x})^2 / (n - 1)$
Unknown-mean variance, raw observations	$[\Sigma x_i^2 - n(\bar{x})^2] / (n - 1)$
Unknown-mean variance, frequency data	$[\Sigma f_i x_i^2 - n(\bar{x})^2] / (n - 1)$

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